

Data Quality Report – E-Commerce Returns Dataset

Dataset: [link to data](#)

Notebook: [link to notebook](#)

Dataset Overview

Property	Value
Raw rows	20,000
Columns	16
Target variable	returned (Yes/No)
Rows after cleaning	~19,940

Dataset Columns

- **Order Information:** order_id
 - **Customer Information:** customer_id, customer_age, customer_gender, past_return_rate
 - **Product Information:** product_category, product_price, order_quantity, discount_percent
 - **Transaction Information:** payment_method, shipping_method
 - **Delivery Information:** delivery_time_days, delivery_delay_days
 - **Customer Feedback:** customer_rating, return_reason
 - **Target Variable:** returned
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Data Quality Assessment

1. Duplicate Records

Issue: The dataset contained 40 exact duplicate rows and 40 duplicate order_ids with minor inconsistencies in fields such as delivery_time_days and customer_rating.

Action Taken: Removed exact duplicates using drop_duplicates() and retained the first occurrence of each duplicate order_id.

2. Product Category Standardization

Issue: The product_category column contained inconsistent labels due to variations in capitalization, spacing, and naming.

Action Taken: Standardized all values into seven canonical categories using a mapping dictionary.

3. Customer Gender Standardization

Issue: The customer_gender column contained inconsistent labels, abbreviations, extra spaces, and missing values.

Action Taken: Standardized all values into Female, Male, Other, and Unknown, with missing values assigned to Unknown.

4. Payment Method Standardization

Issue: Payment methods had inconsistent naming conventions and 199 missing values.

Action Taken: Standardized all payment method labels and replaced missing values with Unknown.

5. Return Status Consistency

Issue: Logical inconsistencies existed between returned and return_reason.

Action Taken: Assigned "Not Returned" when returned = No and "Not Specified" for missing return reasons when returned = Yes, eliminating all inconsistencies.

6. Customer Age Validation

Issue: Customer ages outside the valid range (18–80 years) were identified.

Action Taken: Replaced invalid values with missing values, imputed them using the median age, and rounded the results to integers.

7. Customer Rating Validation

Issue: A total of 677 customer ratings were missing or outside the valid range of 1–5.

Action Taken: Invalid ratings were imputed using category-wise median values, with the global median used as a fallback.

8. Discount Validation

Issue: 80 discount values were outside the valid range of 0–100%.

Action Taken: Clipped all discount values to the valid range while preserving the affected records.

9. Delivery Data Validation

Issue: Missing delivery times, invalid delivery durations, early deliveries, and extreme delivery delays were identified.

Action Taken: Reconstructed missing delivery times, retained valid early deliveries, and flagged delays exceeding 10 days using the `is_extreme_delivery_delay` indicator.

10. Missing Customer Information

Issue: Missing values were found in customer-related attributes, primarily `past_return_rate`.

Action Taken: Missing values were first filled using other records from the same customer and then imputed using global median or mode values where necessary.

11. Missing Transaction Information

Issue: Missing values existed in transaction-related variables before earlier cleaning steps.

Action Taken: After data cleaning, all transaction fields were fully populated with no remaining missing values.

12. Outlier Detection

Issue: Significant outliers were detected in variables such as `product_price`, `order_quantity`, `delivery_time_days`, and `delivery_delay_days`.

Action Taken: Outliers were retained, while capped versions of key variables, log-transformed features, and business-specific indicator variables were created to support robust analysis and modeling.

Data Cleaning Summary

Issue	Records Affected	Resolution
Exact duplicate rows	40	Removed
Duplicate order IDs	40	Deduplicated
Inconsistent product categories	~195	Standardized
Inconsistent customer gender	~935	Standardized
Inconsistent payment methods	~400	Standardized
Returned/Reason conflicts	290	Corrected
Invalid customer ages	~200	Imputed
Invalid customer ratings	677	Imputed
Invalid discount values	80	Clipped
Delivery-related issues	359	Reconstructed and flagged
Missing customer fields	199	Imputed
Missing transaction fields	Fully resolved	Completed
Outliers	1,680+	Flagged and capped
